Data Management and Access Control

Over the past **10 years, ELIXIR-CZ** has become a key player in the **data management and access control**, supporting the Czech and international scientific communities. This **cross-domain strategic area** ensures **secure, efficient, and FAIR (Findable, Accessible, Interoperable, and Reusable) data handling** across disciplines. Over time, our efforts have granularized into **significant services provided from CZ to ELIXIR and beyond**, advancing **authentication, authorization, and access control** for research infrastructures. Aligned with global standards and integrated with national and European projects, ELIXIR-CZ strengthens **data interoperability, security, and accessibility**, empowering researchers with **cutting-edge tools and training** to drive impactful scientific discoveries.

Authentication & Access Control for Life Sciences

ELIXIR-CZ led the development of AAI, which evolved and now serves a broad range of infrastructures.

ELIXIR-CZ has been at the **forefront of authentication and access control** development, together with ELIXIR FI driving the creation of **ELIXIR AAI**, which later evolved into the broader **Life Science AAI (LS AAI)**. This infrastructure now supports not only ELIXIR but also BBMRI or Euro-BioImaging, with services from INSTRUCT, EATRIS, and other research infrastructures. Our contributions have focused on enhancing user-centric access through innovations like GA4GH passports and visas, enabling seamless and flexible delegation-based access control. As part of our continued efforts, we are actively working on **expanding LS AAI to meet the needs of sensitive human data** processing, aligning with initiatives such as EOSC ENTRUST, GDI, and AARC TREE. These advancements are crucial in ensuring that biomedical and life science data remains both **secure and accessible**, fostering greater **interoperability across European and global research infrastructures**.

Secure & Scalable Access to Sensitive and Big Data

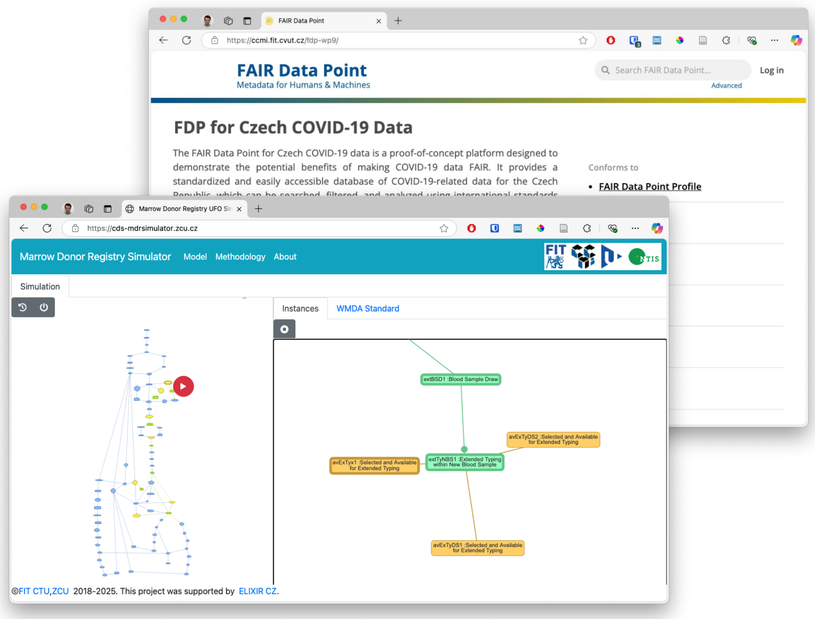
ELIXIR-CZ plays a critical role in advancing AAI for secure, large-scale biomedical data access.

One of our focuses also lies in **enhancing secure access to sensitive and big data**, ensuring that researchers can store, share, and analyze data under strict privacy and security standards. A major achievement has been the transition of the **SensitiveCloud** to full operational status, now holding **over 1.7 PB of data**, making it a crucial resource for handling sensitive biomedical datasets. In parallel, we have successfully set up a **Federated EGA** (FEGA) node at Masaryk University, to which major institutions have committed to join. Additionally, the **ISO 27000 certification** of the SensitiveCloud and AAI team (CESNET and CERIT-SC/MU) demonstrates our commitment to high-security standards. ELIXIR-CZ continues to play a **key role in European projects** like GDI and EOSC-ENTRUST, as well as a strong involvement in EOSC implementation at the national and international levels, ensuring that secure, scalable, and flexible access control solutions are available to researchers across disciplines.

Advancing FAIR Data & Semantic Interoperability

ELIXIR-CZ drives FAIRification efforts, ensuring that data across diverse research domains is FAIR.

ELIXIR-CZ is a **key advocate for FAIR data principles** in life sciences, actively shaping discussions on FAIR data implementation within EOSC CZ, particularly in the Bio/Med/Food research domains. As part of the ELIXIR-CONVERGE project, we delivered one of the first **FAIRification use cases,** focusing on COVID-19 datasets, setting an important precedent for making research data more structured, accessible, and reusable. A major contribution to this effort is the Data Stewardship Wizard (DSW), which has been recognized as an ELIXIR Recommended Interoperability Resource. The **DSW supports FAIR data from the planning stage**, helping researchers design FAIR-compliant **data management plans (DMPs)** and develop **FAIR Implementation Profiles** in collaboration with the **GO-FAIR Foundation**. These efforts improve the interoperability of historically fragmented data collections, ensuring that research data can be effectively shared and reused across disciplines and infrastructures. Various projects contributed to **semantic interoperability**, such as **Marrow Donor Registry Simulator.**



Building a Comprehensive Data Management Toolkit

ELIXIR-CZ has developed a robust toolkit supporting researchers throughout the entire FAIR data lifecycle.

We have been actively developing a comprehensive Data Management (DM) toolkit that supports researchers throughout the entire FAIR data lifecycle—from data capture and annotation to integration, sharing, and publication. A key component of this toolkit is the **Data Stewardship Wizard (DSW)**, which has matured significantly and achieved **high adoption across Europe and beyond**. DSW supports **machine-actionable Data Management Plans (maDMPs)**, making best-practices-aligned data management more efficient and structured. The tool has also become an **integral part of the FAIR-tools ecosystem**, featuring integrations with RDMkit, FAIR Cookbook, DAISY/DPIA, RO-Crate, and Galaxy, ensuring seamless workflows across research platforms. Progress in this area is driven by **strong collaborations within ELIXIR**, particularly through BioHackathons, ELIXIR-CONVERGE, and ELIXIR-STEERS, as well as international initiatives like RDA and EOSC. Through these efforts, ELIXIR-CZ provides researchers with the necessary **tools and frameworks to ensure their data is FAIR, well-managed, and ready for future reuse**.

Empowering Researchers & Data Stewards by Training

ELIXIR-CZ strengthens research capabilities through providing training in AAI, RDM, and FAIR principles.

Training is a cornerstone of our strategy, ensuring that researchers are equipped with key skills in **AAI, data management planning, and FAIR data principles**. Our training programs are **accessible through ELIXIR TeSS**, offering a broad range of resources for scientists at all levels. Recently, we have placed a strong emphasis on **train-the-trainer initiatives**, expanding the network of skilled professionals who can further disseminate best practices. Additionally, we focus on developing **FAIR training materials**, helping researchers seamlessly integrate FAIR principles into their workflows. By fostering a culture of knowledge-sharing, ELIXIR-CZ supports the long-term sustainability of FAIR data management across disciplines and throughout various initiatives beyond ELIXIR (such as EOSC).

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